SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: **ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS** **COURSE CODE:** **MLA0104**

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50 **Time: 60 Mins**

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I M . M E G H A N A D H (1 9 2 3 2 5 0 2 6) __ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.MEGHANADH

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty Total Marks Awarded

Introduction and Background

Artificial Intelligence is transforming modern manufacturing by making production smarter, faster, and more reliable. One of the most important AI applications is Predictive Maintenance. Instead of waiting for machines to fail, AI predicts failures before they occur. This helps industries reduce downtime, improve productivity, and lower maintenance costs. AI uses data collected from sensors installed on machines to monitor their condition continuously. By analysing this data, AI can identify unusual patterns and alert engineers before a machine breaks down.

Problem Identification

Traditional maintenance methods either repair machines after they fail or perform maintenance at fixed intervals. These methods are expensive and often result in unexpected breakdowns, production delays, and higher repair costs. Manufacturing industries need a smarter solution that predicts failures in advance and ensures uninterrupted production.

Need for AI Solution

AI can process large amounts of machine data in real time. It detects hidden patterns that humans cannot easily identify. Predictive maintenance reduces downtime, increases equipment life, improves product quality, minimizes maintenance costs, and enhances worker safety.

AI Technologies Used

- Machine Learning
- Deep Learning
- Internet of Things (IoT)
- Computer Vision
- Big Data Analytics
- Cloud Computing

Working Mechanism

Sensors collect information such as temperature, vibration, pressure, and sound from industrial machines. The data is sent to cloud servers where AI algorithms analyse it. The system compares current data with historical records to identify abnormal behaviour. If a possible fault is detected, the AI system sends an alert so maintenance can be performed before the machine fails.

Data Sources and Processing

The system uses sensor data, maintenance records, production logs, and machine performance history. The collected data is cleaned, processed, and analysed using machine learning algorithms to predict equipment health and future failures.

Real-World Implementation

Many companies use AI-based predictive maintenance, including:

- Siemens
- General Electric (GE)
- Bosch
- BMW
- Tata Steel
- Toyota

These organizations use AI to improve machine reliability and reduce operational costs.

Impact and Benefits

- Reduces machine downtime
- Improves production efficiency
- Lowers maintenance costs
- Extends equipment lifespan
- Improves product quality
- Increases workplace safety
- Saves energy and resources

Challenges and Limitations

- High installation cost
- Large amounts of quality data required
- Cybersecurity risks
- Complex AI model development
- Need for skilled professionals
- Integration with old manufacturing systems

Future Scope and Emerging Trends

Future AI systems will use Digital Twins, Edge AI, 5G communication, autonomous robots, and Generative AI. Smart factories will become more intelligent, self-monitoring, and capable of making automatic maintenance decisions with minimal human intervention.

Conclusion

AI-based Predictive Maintenance is one of the most important technologies in Industry 4.0. It helps manufacturers reduce costs, prevent unexpected failures, improve productivity, and increase equipment reliability. As AI technology continues to evolve, predictive maintenance will play a vital role in building smarter, safer, and more efficient manufacturing industries.

References

1. Siemens – Smart Manufacturing Solutions
2. General Electric Digital – Predix Platform
3. IBM – AI in Manufacturing
4. Microsoft Azure AI for Manufacturing
5. Google Cloud Manufacturing Solutions
6. IEEE Xplore Research Papers
7. Springer – AI in Smart Manufacturing
8. Elsevier – Journal of Manufacturing Systems
9. McKinsey & Company – AI in Manufacturing
10. Deloitte – Smart Factory Reports

This content matches the report structure required in your assessment for the **Manufacturing** domain.